

AI-Driven Intelligent Security Systems for Multi-Domain Threat Detection and Response

Capstone Group Research Report (Draft v2.0) — Group 5

Integrating four individual projects: Phishing Detection (Nguyễn Ngọc Bách), Behavior-Based Continuous Authentication & Web Attack Detection (Trần Văn Khôi), Multimodal Deepfake Detection (Phạm Quang Huy), and SentinelAI SOC Agent (Nguyễn Đức Bình)

Version: Draft v2.0 (proposal/report ready for Google Docs / PDF export)

Date: 2026-03-11 (Asia/Bangkok)

Executive Summary

Security teams increasingly defend against threats that span multiple domains: social engineering (phishing emails and fake websites), malicious web requests (SQLi/XSS/LFI/RFI and related injection patterns), identity/session compromise (credential theft followed by session hijack), and media manipulation (deepfakes used for fraud and impersonation). At the same time, SOCs face alert overload and “alert fatigue,” which can delay incident detection and response. ¹

This capstone proposes a **multi-domain, AI-driven security ecosystem** that provides detection signals from four specialized modules and unifies them into a **single threat intelligence and response layer**:

- **Module A — Phishing Detection (Bách):** Detect phishing emails and fake websites using NLP + ML, leveraging authoritative phishing intelligence feeds and curated corpora (e.g., PhishTank, APWG) and realistic benign email corpora (Enron). ²
- **Module B — Behavior-Based Continuous Authentication & Web Attack Detection (Khôi):**
 - Continuous authentication using behavioral biometrics (keystroke/mouse/touch/sensor signals) with robust evaluation metrics (FAR/FRR/EER). ³
 - Web attack detection from HTTP requests using NLP-inspired tokenization and deep models on public web-attack datasets (CSIC 2010, CAPEC-labeled datasets). ⁴
- **Module C — Multimodal Deepfake Detection (Huy):** Detect deepfakes with video-only and audio-visual consistency features, emphasizing robustness to manipulation methods and post-processing (compression). ⁵
- **Module D — SentinelAI SOC Agent (Bình):** An AI-assisted SOC decision-support “agent” that ingests security telemetry and module outputs, correlates alerts into incidents, maps/labels behaviors using MITRE ATT&CK where feasible, and recommends response actions aligned with NIST incident response guidance (human-in-the-loop). ⁶

Core novelty (group-level): Instead of building four unrelated prototypes, the group builds **four vertical detectors + one horizontal integration layer** that normalizes events (OCSF-style mapping), unifies risk

scoring, and supports SOC-aligned workflows (detect → respond → recover) with auditability and safeguards. ⁷

Primary outcomes: (1) a cohesive architecture and shared data model; (2) four module prototypes (each runnable on public datasets); (3) an integration demo showing cross-domain correlation; and (4) a measurement-driven evaluation with both **module-level metrics** (precision/recall/F1/AUC/EER) and **SOC-level metrics** (false positive reduction, alert reduction rate, time-to-triage proxy, and response recommendation quality). ⁸

Introduction and Problem Context

Context and motivation

Organizations are attacked through multiple entry points and modalities. Phishing remains highly active and continuously evolving (including trends such as QR-code-driven phishing and mixed-media social engineering), requiring detection that understands both **textual intent** and **link/brand deception behaviors**. ⁹

Web applications remain frequent targets via malicious HTTP requests (e.g., injection payloads), which can be expressed as structured text-like sequences (URLs, parameters, headers), enabling NLP-style modeling but also creating dataset realism and generalization challenges. ¹⁰

Simultaneously, attackers increasingly exploit identity: compromised credentials and session hijacking may not be prevented by one-time login; thus continuous authentication (behavioral biometrics) provides an additional layer to detect “impostor drift” during sessions. ¹¹

Deepfakes add a new deception vector, enabling impersonation, fraud, and social engineering at scale; robust detection must generalize across manipulation methods and withstand compression and distribution artifacts. ¹²

Finally, SOC must operationalize detection into incident response. Modern incident response guidance emphasizes integration with risk management and continuous improvement; responders need triage, correlation, and response recommendations that reduce overload without unsafe automation. ¹³

Problem statements (group-level)

PS1 (Siloed detection): Many AI security works focus on a single threat class; SOC teams, however, must handle **multi-domain signals** and correlate them into incidents. ¹⁴

PS2 (Generalization and robustness): Models trained on a dataset frequently degrade under distribution shift: new phishing templates, new web attacks, unseen deepfake methods, and behavior drift. ¹⁵

PS3 (Operational constraints): Even accurate detectors can be operationally unusable if they produce high false positives, require heavy compute, or cannot provide explainability/audit trails. ¹⁶

PS4 (Response safety): Autonomous response without safeguards can cause service disruption. Response must be policy-based, logged, and human-approved for high-impact actions. ¹⁷

Scope

Included: building four module prototypes + one integration layer (shared schema, shared risk scoring, SOC-oriented correlation) evaluated on public datasets and simulated/synthetic SOC telemetry. ¹⁸

Excluded: production deployment at enterprise scale; offensive exploitation; real-time enforcement in a live national infrastructure environment (the capstone is a prototype + evaluation).

SMART objectives

The group commits to objectives that are **Specific, Measurable, Achievable, Relevant, and Time-bound** within the capstone duration:

- **O1 (Phishing module performance):** Achieve $F1 \geq 0.90$ on at least one public phishing dataset split (email or URL/website) and report precision/recall at operationally meaningful thresholds. ¹⁹
- **O2 (Web attack module performance):** Achieve $F1 \geq 0.95$ on CSIC 2010 or CAPEC-labeled datasets with explicit reporting of false positive rate at a chosen operating point. ²⁰
- **O3 (Continuous authentication):** Implement a behavioral authentication prototype that reports **FAR/FRR/EER**, and demonstrate stable user verification in at least one public dataset (keystroke/touch/mouse). ²¹
- **O4 (Deepfake detection):** Evaluate on at least **two deepfake benchmarks** (e.g., FaceForensics++ and one of DFDC/Celeb-DF) and report AUC/F1 plus cross-dataset generalization. ²²
- **O5 (Integrated SOC demo):** Demonstrate a working pipeline where module outputs are normalized and fused into an incident view, and show at least **one “multi-signal incident scenario”** (e.g., phishing → anomalous login behavior → suspicious web requests) using simulated telemetry. ²³

Research questions

RQ1: Which AI-based methods are most effective for each sub-domain (phishing, web attacks, continuous authentication, deepfake detection), and what are their strengths/limitations in generalization and operational use? ²⁴

RQ2: How can outputs from heterogeneous detectors be normalized (schema), correlated (incident formation), and prioritized (risk scoring) into SOC workflows? ²⁵

RQ3: How should the integrated system be evaluated with both AI metrics (precision/recall/F1/AUC/EER) and SOC metrics (alert reduction, time-to-triage proxy, response safety/usability)? ²⁶

Literature Review

This literature review is organized by **the four project domains plus cross-cutting standards**. For each subtopic, we highlight 5–8 core sources (papers/datasets/official guidance) and extract what matters for the capstone: datasets, modeling choices, and evaluation pitfalls.

Phishing detection (email + fake websites) — key sources (5–8)

- 1) **APWG Phishing Activity Trends Reports** document the continuing scale and evolution of phishing, supporting urgency and realistic threat framing. ⁹
- 2) **PhishTank** provides a collaborative platform and an open API for phishing site intelligence—useful for building URL/website ground truth pipelines. ²⁷
- 3) **Enron Email Dataset** provides a large benign email corpus for training and evaluation in email-based classification settings. ²⁸
- 4) **CANTINA+ (ACM)** presents a layered ML-based approach for phishing website detection, demonstrating feature-rich detection and operational focus on false positives. ²⁹
- 5) **BERT-PhishFinder (IEEE Computer Society)** exemplifies transformer-based phishing detection for improved robustness in text classification. ³⁰
- 6) **RNN-based phishing email detection** (Halgas, 2020) shows deep sequence modeling for email detection using Enron + phishing corpora. ³¹
- 7) **Phishing website datasets (Data in Brief)** (Vrbančič et al.) provide large labeled phishing/benign website datasets suitable for ML training. ³²
- 8) **PhreshPhish (2025)** proposes a large-scale “real-world” phishing URL dataset collection methodology and highlights collection challenges. ³³

Main gaps relevant to this capstone: (a) models often overfit to narrow template artifacts; (b) many works do not report operational FPR at realistic thresholds; (c) integration with SOC workflows (triage, escalation) is frequently missing. ³⁴

Behavior-based continuous authentication — key sources (5–8)

- 1) **Continuous authentication behavioral biometrics review (ACM, 2016)** provides a foundational survey of CA methods and design considerations. ³⁵
- 2) **Eberz et al. (AsiaCCS, 2017)** focuses on CA evaluation pitfalls and emphasizes robust metrics beyond averaged FAR/FRR. ³⁶
- 3) **Killourhy & Maxion (DSN 2009)** provide a keystroke benchmark dataset and repeatable evaluation procedure, critical for reproducible CA testing. ³⁷
- 4) **Balabit Mouse Dynamics Challenge dataset** enables standardized mouse-based authentication evaluation. ³⁸
- 5) **Touchalytics (2012)** demonstrates continuous authentication via touchscreen features and reports EER over time. ³⁹
- 6) **HMOG dataset (Sitová et al.)** supports multimodal behavior signals for CA. ⁴⁰
- 7) **BehavePassDB (Pattern Recognition, 2023)** provides a modern benchmark for mobile behavioral biometrics and evaluation. ⁴¹

Main gaps relevant to this capstone: (a) CA performance may collapse under device changes/time drift; (b) evaluation must model realistic impostors; (c) deployment must respect privacy and avoid capturing sensitive content (only timing/behavioral features). ⁴²

Web attack detection from HTTP requests (NLP-style) — key sources (5–8)

- 1) **CSIC 2010 HTTP dataset** is a canonical public dataset for HTTP request anomaly/attack detection, though it is synthetic and may not represent modern attack diversity. ⁴³

- 2) **Liang et al. (ACM, 2017)** propose deep learning for anomaly-based web attack detection, motivating representation learning over hand-crafted rules. ⁴⁴
- 3) **Wang et al. (ACM, 2018)** evaluate CNN and LSTM for web attack detection, informing architecture choices for sequence-like payloads. ⁴⁵
- 4) **Tian et al. (IEEE TII)** propose distributed deep learning for web attack detection and supplies a key reference for scalable architectures. ⁴⁶
- 5) **Tekerek (Computers & Security, 2021)** proposes CNN-based web attack detection on web query representations (relevant for practical modeling choices). ⁴⁷
- 6) **Riera et al. (Computers & Security, 2022)** propose a CAPEC-based multi-label dataset and classification approach, enabling “attack type” labeling instead of binary only. ⁴⁸
- 7) **ATT&CK technique data in STIX** provides a structured way to map certain web attacks (e.g., exploit public-facing app) to standardized tactics/techniques for SOC reporting. ⁴⁹

Main gaps relevant to this capstone: (a) many datasets are synthetic/old, harming generalization; (b) multi-label taxonomy (attack categories) is often missing; (c) operational integration (WAF/SIEM correlation) is under-specified. ⁵⁰

Multimodal deepfake detection — key sources (5–8)

- 1) **FaceForensics++ (ICCV 2019)** is a cornerstone benchmark for manipulated face detection and highlights compression robustness challenges. ⁵¹
- 2) **DFDC dataset (arXiv + Meta dataset page)** provides very large-scale face swap video data designed for detection progress measurement. ⁵²
- 3) **Celeb-DF (CVPR 2020)** provides higher-quality deepfakes that are more representative of “in-the-wild” content than early datasets. ⁵³
- 4) **DeeperForensics-1.0 (CVPR 2020)** adds scale and real-world perturbations, stressing generalization under distortions. ⁵⁴
- 5) **LipForensics (CVPR 2021)** targets semantic mouth-motion irregularities, aiming to generalize to unseen manipulations and resist compression. ⁵⁵
- 6) **Audio-video sync foundations (Chung & Zisserman, 2016)** motivate using learned audio-visual synchronization as a fraud/deepfake cue. ⁵⁶
- 7) **Lip-sync deepfake detection from audio-video mismatch (CVPRW 2024)** motivates specialized detectors for “lip-sync-only” manipulations. ⁵⁷

Main gaps relevant to this capstone: (a) cross-dataset performance often drops; (b) detectors can be brittle to compression; (c) multimodal fusion must be designed to avoid overfitting to dataset-specific artifacts. ⁵⁸

SentinelAI SOC agent and cross-cutting standards — key sources (5–8)

- 1) **NIST SP 800-61r3 (2025)** updates incident response guidance and emphasizes integration across CSF 2.0 functions; supports safe response design with continuous improvement. ⁵⁹
- 2) **NIST SP 800-92 (Log Management)** supports design choices for telemetry collection, normalization, retention, and analysis. ⁶⁰
- 3) **NIST SP 800-137 (ISCM)** frames continuous monitoring as ongoing risk awareness, grounding SentinelAI’s monitoring purposes. ⁶¹
- 4) **MITRE ATT&CK Design & Philosophy** and **ATT&CK resources** provide standardized adversary behavior

modeling, enabling consistent incident labeling and detection validation. ⁶²

5) **SoK: MITRE ATT&CK in Research and Practice (2023)** highlights gaps in ATT&CK evaluation/usage—useful for realistic integration planning. ⁶³

6) **Alert Fatigue in SOCs (ACM)** and **AlertPro (Computers & Security)** motivate alert prioritization/correlation as operational necessities. ⁶⁴

7) **MITRE Caldera** and **Splunk Attack Range** enable safe attack simulation and telemetry generation for evaluation in lab environments. ⁶⁵

8) **OCSF (Open Cybersecurity Schema Framework)** supports vendor-agnostic normalization across heterogeneous signals. ⁶⁶

Proposed Solution and Architecture

System concept

Group 5 will deliver a **modular ecosystem**:

- Four “domain detectors” produce **structured security findings** (alerts + evidence + confidence).
- A **shared normalization** layer maps outputs into a common schema (OCSF-inspired) and enriches with threat intel (ATT&CK mapping where appropriate). ⁶⁷
- A **correlation and decision layer** (SentinelAI) groups events into incidents, ranks risk, and recommends response actions aligned with incident response guidance and human oversight. ⁶⁸

Professional diagram (reference model)

The project aligns its response workflow to **NIST SP 800-61r3** incident response lifecycle framing (CSF 2.0 oriented), emphasizing Detect → Respond → Recover and continuous improvement. ⁶⁹

(In the Google Doc, insert the figure with a caption: “Incident response life cycle model (NIST SP 800-61r3)” and cite NIST.) ⁷⁰

Overall ecosystem architecture (Mermaid flowchart)

```
flowchart LR
  subgraph Sources["Security & Content Data Sources"]
    EML["Email content + headers"]
    URL["URLs + Web page content/DOM"]
    HTTP["HTTP requests (URL, params, headers, body)"]
    BIO["Behavioral biometrics (keystroke/mouse/touch/sensors)"]
    VID["Video/audio streams (media authenticity)"]
    SOC["SOC telemetry (EDR, auth logs, DNS, proxy, SIEM events)"]
  end

  subgraph Ingest["Ingestion & Normalization Layer"]
    COL["Collectors/Connectors"]
    NORM["Parsing + Normalization (OCSF-like event mapping)"]
  end
```

```

    ENR["Enrichment (WHOIS, URL reputation, ATT&CK tags, geo, asset context)"]
end

subgraph Modules["AI Detection Modules (Group 5)"]
    PHISH["Module A: Phishing Detection (NLP/ML)"]
    AUTH["Module B1: Continuous Authentication (Behavior ML)"]
    WEBATK["Module B2: Web Attack Detection (NLP on HTTP)"]
    DEEP["Module C: Multimodal Deepfake Detection (AV + biometrics)"]
end

subgraph Fusion["Fusion, Correlation & Response"]
    AGENT["Module D: SentinelAI SOC Agent\nCorrelation + Risk Scoring + Playbooks"]
    CASE["Case/Incident Store"]
    DASH["SOC Analyst UI (triage, evidence, actions)"]
    SOAR["SOAR/Automation Hooks (guardrailed)"]
end

Sources --> COL --> NORM --> ENR
ENR --> PHISH
ENR --> AUTH
ENR --> WEBATK
ENR --> DEEP

PHISH --> AGENT
AUTH --> AGENT
WEBATK --> AGENT
DEEP --> AGENT
SOC --> COL

AGENT --> CASE --> DASH
AGENT --> SOAR

```

Per-module architectures (high-level pipelines)

Module A — Phishing detection (email + fake websites)

```

flowchart TB
    A1["Email/Webpage Input"] --> A2["Preprocess\n- decode HTML\n- extract headers\n- extract URLs/domains\n- clean text"]
    A2 --> A3["Feature & Representation\n- URL lexical + host features\n- text embeddings (TF-IDF / Transformer)"]
    A3 --> A4["Model\n- baseline ML (LR/RF/XGBoost)\n- Transformer classifier (BERT-family)"]

```

```
A4 --> A5["Output\n- phishing probability\n- evidence (salient terms/URLs)\n- recommended action"]
```

Data sources and ground truth pipelines will be built using PhishTank/APWG feeds and public corpora, pairing with benign corpora like Enron to create train/test splits. ⁷¹

Module B1 — Continuous authentication (behavioral biometrics)

```
flowchart TB
    B1["User Interaction Events\n(keystroke/mouse/touch/sensors)"] --> B2["Feature extraction\n- timing stats\n- trajectories\n- rhythm vectors\n- session windows"]
    B2 --> B3["Model\n- classifier (RF/XGBoost)\n- sequence model (RNN/TCN)\n- embedding + threshold"]
    B3 --> B4["Decision\n- FAR/FRR tradeoff\n- EER operating point\n- time-to-detect impostor"]
    B4 --> B5["Output\n- confidence of current user\n- session risk signal to SentinelAI"]
```

Evaluation follows best-practice metrics and considers systematic error distributions and attacker models. ⁷²

Module B2 — Web attack detection (NLP on HTTP requests)

```
flowchart TB
    C1["HTTP Request"] --> C2["Canonicalization\n- normalize encoding\n- tokenize URL/params\n- header parsing\n- optional body parsing"]
    C2 --> C3["Representation\n- char/BPE tokens\n- n-grams\n- embeddings"]
    C3 --> C4["Model\n- CNN/LSTM/CNN-LSTM\n- Transformer-based classifier\n- anomaly model (autoencoder)"]
    C4 --> C5["Output\n- attack class (binary or multi-label)\n- confidence\n- extracted indicators"]
```

Benchmarks use established public datasets, including CSIC and CAPEC-labeled multi-label datasets when available. ⁷³

Module C — Deepfake detection (multimodal)

```
flowchart TB
    D1["Video + Audio"] --> D2["Preprocess\n- face detection & alignment\n- frame sampling\n- audio feature extraction (MFCC/SSL embeddings)"]
    D2 --> D3["Visual Track\n- CNN/ViT features\n- temporal model (3D CNN/Transformer)"]
    D2 --> D4["Audio-Visual Consistency\n- lip-sync embeddings (SyncNet-style)\n- "
```



```

cross-modal similarity"]
D3 --> D5["Fusion\n- late fusion (stacking)\n- attention-based fusion"]
D4 --> D5
D5 --> D6["Output\n- deepfake probability\n- robustness flags\n- explanation
(region/temporal cues)"]

```

The design explicitly targets robustness and generalization across datasets and manipulations. 74

Shared data model (Mermaid ER diagram)

This ER model supports unified storage of events, alerts, incidents, and feedback across modules.

```

erDiagram
    EVENT ||--o{ ALERT : produces
    INCIDENT ||--o{ ALERT : contains
    INCIDENT ||--o{ RESPONSE_ACTION : triggers
    INCIDENT ||--o{ EVIDENCE : has
    MODEL_VERSION ||--o{ ALERT : generated_by
    DATASET ||--o{ MODEL_VERSION : trained_on
    ANALYST ||--o{ FEEDBACK : submits
    ALERT ||--o{ FEEDBACK : receives
    ENTITY ||--o{ EVENT : involved_in
    ENTITY ||--o{ INCIDENT : referenced_in

    EVENT {
        string event_id
        datetime ts
        string source_type
        string raw_ref
        string normalized_json
    }
    ALERT {
        string alert_id
        string event_id
        string module_name
        float score
        string label
        json explanation
    }
    INCIDENT {
        string incident_id
        datetime created_ts
        string status
        float risk_score
        json attack_mapping
    }

```

```
RESPONSE_ACTION {
  string action_id
  string incident_id
  string action_type
  string approval_state
  datetime executed_ts
}
EVIDENCE {
  string evidence_id
  string incident_id
  string artifact_type
  string artifact_ref
}
MODEL_VERSION {
  string model_id
  string module_name
  string version
  string hash
}
DATASET {
  string dataset_id
  string name
  string license
}
ANALYST {
  string analyst_id
  string role
}
FEEDBACK {
  string feedback_id
  string alert_id
  string analyst_id
  string outcome_label
  string notes
}
ENTITY {
  string entity_id
  string entity_type
  string name_hash
}
```

Methodology and Implementation Plan

Research methodology: Design Science Research + iterative AI lifecycle

This capstone follows **Design Science Research**: build an artifact (the ecosystem), evaluate it, and iterate based on evidence—aligned with the core principles that knowledge is gained through building and evaluating artifacts. ⁷⁵

DSRM phases applied: - **Problem identification & motivation:** multi-domain threats + SOC operational friction (alert fatigue, correlation gaps). ⁷⁶

- **Define objectives:** SMART objectives O1–O5 (above).

- **Design & development:** modular detectors + normalization + SentinelAI correlation.

- **Demonstration:** run module demos on public datasets; run a multi-signal scenario demo on simulated SOC telemetry. ⁷⁷

- **Evaluation:** metrics-driven benchmarking + ablation tests + integration evaluation. ⁷⁸

- **Communication:** final report + slides + recorded demo.

AI lifecycle and governance:

We incorporate trust and risk management concerns (privacy, harmful failures, monitoring drift) consistent with AI risk management principles. ⁷⁹

Implementation plan (work breakdown)

Group-level milestones: - **M1:** Shared schema + storage + baseline pipeline skeleton (week 1–3)

- **M2:** Each module baseline model trained + evaluated on at least one dataset (week 4–7)

- **M3:** Improved models + robustness tests + cross-dataset tests where feasible (week 8–11)

- **M4:** SentinelAI correlation + incident view + response recommendations (week 8–12)

- **M5:** System integration demo + evaluation report + slides (week 13–16)

Task allocation (ownership matrix):

Workstream	Key tasks	Owner
Shared platform	Event schema mapping, storage (PostgreSQL/Elastic), API conventions, evaluation harness	Whole group (lead: Bình)
Phishing	Data ingestion (PhishTank/APWG + corpora), NLP pipeline, BERT + baseline ML, evaluation & error analysis	Bách
Continuous authentication	Dataset selection, feature extraction windows, FAR/FRR/EER evaluation, drift considerations	Khôi
Web attack detection	HTTP canonicalization/tokenization, deep model training, multi-label experiments if feasible	Khôi
Deepfake	Dataset pipeline (FaceForensics++ + DFDC/Celeb-DF), multimodal features, robustness tests	Huy

Workstream	Key tasks	Owner
SentinelAI	Correlation logic, risk scoring, ATT&CK mapping integration, human-in-loop response	Binh
Reporting	Drafting report, diagrams, tables, slides, references	Whole group

Timeline (Mermaid Gantt)

```

gantt
    title Group 5 Capstone Draft v2.0 Plan (16 weeks)
    dateFormat YYYY-MM-DD
    axisFormat %m/%d

    section Shared Platform
    Schema + data model + APIs           :a1, 2026-03-11, 14d
    Storage + ingestion pipeline         :a2, after a1, 21d
    Integration harness + CI            :a3, after a2, 21d

    section Module A (Phishing)
    Dataset pipelines + preprocessing   :b1, 2026-03-18, 21d
    Baseline ML + BERT training         :b2, after b1, 21d
    Evaluation + error analysis         :b3, after b2, 14d

    section Module B (Auth + Web Attacks)
    CA datasets + feature extraction     :c1, 2026-03-18, 21d
    CA modeling + metrics               :c2, after c1, 21d
    HTTP attack dataset + tokenization  :c3, 2026-03-25, 21d
    Web attack models + evaluation       :c4, after c3, 21d

    section Module C (Deepfake)
    Deepfake datasets + preprocessing   :d1, 2026-03-18, 21d
    Visual + AV models                 :d2, after d1, 28d
    Cross-dataset testing               :d3, after d2, 14d

    section Module D (SentinelAI)
    Correlation + risk scoring          :e1, 2026-04-15, 28d
    ATT&CK mapping + response playbooks :e2, after e1, 14d
    Integrated demo                    :e3, after e2, 14d

    section Finalization
    Evaluation write-up + figures       :f1, 2026-06-01, 10d
    Slides + rehearsal                 :f2, after f1, 7d

```

Datasets, Models, and Experiment Design

Dataset selection table (cross-module)

The table below lists candidate datasets prioritized for public availability and research usage.

Domain	Dataset	What it contains	Why it's relevant	Known limitations
Phishing (URLs)	PhishTank feed/API ²⁷	Verified phishing URLs and metadata	Real-world phishing intelligence for URL labeling	Feed coverage and label noise depend on submissions
Phishing (email)	Enron Email Dataset ⁸⁰	~0.5M emails from Enron employees	Large benign corpus; useful for ham baseline	Not phishing-focused; requires pairing with phishing corpus
Phishing (websites)	Vrbancič phishing websites datasets ³²	Large labeled phishing vs benign websites	Scalable training/eval for website classifiers	Feature extraction may differ from real-time browsing
Web attacks	CSIC 2010 HTTP dataset ⁴³	Labeled normal vs anomalous HTTP requests	Canonical benchmark for HTTP attack detection	Synthetic and dated; limited modern attacks
Web attacks (multi-label)	CAPEC-labeled dataset (Riera et al., 2022) ⁴⁸	Multi-label web attack types	Enables “attack type” classification	Access method may vary; feature choices affect comparability
Continuous auth (keystroke)	CMU Keystroke benchmark dataset ⁸¹	Repeated password typing timings	Standard for keystroke authentication	“Fixed text” may not match free-text scenarios
Continuous auth (mouse)	Balabit Mouse Dynamics Challenge ³⁸	Mouse pointer timing/position data	Standard mouse biometrics benchmark	Data may not represent modern UI/device diversity
Continuous auth (mobile touch)	Touchalytics ³⁹	Touchscreen gestures for CA	Demonstrates CA feasibility and EER reporting	Performance may degrade over time, per study
Continuous auth (multimodal)	HMOG dataset ⁴⁰	Touch + motion/grasp signals	Multimodal CA evaluation	Smartphone-specific; privacy considerations

Domain	Dataset	What it contains	Why it's relevant	Known limitations
Deepfake	FaceForensics++ ⁸²	Multiple manipulation methods, compression	Core benchmark for face manipulation detection	Domain bias; high dataset familiarity in literature
Deepfake	DFDC dataset ⁵²	Large face-swap dataset	Scale and realism; challenge benchmark	Heavy compute/storage; still domain gaps
Deepfake	Celeb-DF ⁵³	High-quality celebrity deepfakes	More “in-the-wild”-like quality	Focused on celebrity domain
SOC telemetry	CICIDS2017 ⁸³	PCAP + labeled flows with common attacks	Standard IDS benchmark	Network-only perspective, limited endpoint context
SOC telemetry	LANL auth logs ⁸⁴	Large enterprise authentication events	Supports auth anomaly + correlation	Massive scale; not labeled as “attacks” in a simple way
SOC telemetry (synthetic)	CMU SEI Insider Threat Dataset ⁸⁵	Synthetic enterprise logs + malicious actors	Supports user-behavior correlation	Synthetic; assumptions may differ from real environments
SOC telemetry (simulated)	Splunk Attack Range + MITRE Caldera ⁸⁶	Attack emulation + generated telemetry	Creates “controlled ground truth” data	Lab realism depends on configuration

Synthetic / simulated SOC log generation (for integration demos)

To evaluate the integrated system realistically without sensitive operational logs, the project will generate telemetry using widely used adversary emulation tools that explicitly support safe simulations and detection engineering workflows. ⁸⁶

- **Threat simulation and telemetry generation:** MITRE Caldera (ATT&CK-based adversary emulation) and Splunk Attack Range (instrumented environments + attack simulation) enable repeatable scenarios. ⁸⁷
- **Log management principles:** Logging/retention/normalization designs follow NIST guidance for log management and continuous monitoring. ⁸⁸

Models & algorithms (specific, per module)

Module A (Phishing Detection) — candidate models - Baselines: Logistic Regression / Random Forest / Gradient Boosting on engineered features (URL lexical, domain features, header cues).

- **NLP deep models:** Transformer-based classifiers (BERT-family) for email body/subject and website text;

align with modern phishing NLP work. ⁸⁹

- **Hybrid approach:** Combine rule-based indicators (e.g., suspicious URL patterns) with learned models; CANTINA+ supports layered strategies and emphasizes low false positives. ²⁹

Module B1 (Continuous Authentication) — candidate models - Statistical + ML: RF/XGBoost on windowed behavioral features; evaluate FAR/FRR/EER. ⁹⁰

- **Sequence models:** RNN/Temporal ConvNet over time windows for continuous decisions; use datasets like keystroke benchmark, Touchalytics, HMOG. ⁹¹

- **Embedding + threshold:** Learn user embeddings (metric learning) then threshold; calibrate and report operating points carefully due to systematic bias risks. ⁹⁰

Module B2 (Web Attack Detection) — candidate models - Token-based deep learning: char-level CNN, LSTM, CNN-LSTM as indicated by prior evaluations. ⁴⁵

- **Anomaly detection:** autoencoder for “normal-only” training to flag unknown attacks (aligned with anomaly-based detection literature). ⁹²

- **Multi-label classification:** Use CAPEC-labeled datasets as available to provide “attack category” output rather than binary only. ⁹³

Module C (Deepfake Detection) — candidate models - Visual backbone: Xception/EfficientNet/ViT features; temporal modeling for video-level decisions (3D CNN/temporal transformer). ⁸²

- **Mouth-motion semantics:** LipForensics-like mouth embedding + temporal classifier to generalize beyond low-level artifacts. ⁹⁴

- **Audio-visual consistency:** SyncNet-like learned A/V sync cues, augmented by modern AV representations as feasible. ⁹⁵

Module D (SentinelAI SOC Agent) — candidate methods - Normalization and shared schema: OCSF mapping for consistent ingestion across sources and module outputs. ⁹⁶

- **Correlation engine:** graph-based correlation on entities (user, host, IP, domain, URL) + time windows; store as incidents.

- **Risk scoring:** calibrated scoring model (e.g., gradient boosting over alert features + contextual enrichment) combined with rule-based escalation thresholds; incorporate “continuous improvement” feedback loops aligned to incident response guidance. ⁹⁷

- **Threat taxonomy mapping:** use ATT&CK STIX data to map relevant findings into standardized technique tags when applicable. ⁴⁹

Evaluation, Risk, and Ethics

Evaluation metrics (module-level + system-level)

Module-level metrics - Classification modules (phishing/web attacks/deepfake): Precision, Recall, F1, ROC-AUC; additionally report **false positive rate** at chosen operating points to reflect operational constraints. ⁹⁸

- **Continuous authentication:** FAR, FRR, EER; also report time-window sensitivity (how quickly an impostor is detected) and stability over time (session drift). ⁹⁹

System-level SOC metrics - Alert reduction rate: ratio of raw alerts to correlated incidents (goal: fewer, higher-quality incidents).

- **False positive reduction:** how many low-confidence alerts are suppressed or downgraded without missing key events.

- **Time-to-triage proxy / MTTR proxy:** measure the time SentinelAI needs to produce an incident package (evidence + recommendation) compared to a baseline “no-correlation” flow; aligns with the operational goals of incident response improvements. ⁶⁸

- **Usability/analyst utility (lightweight):** structured questionnaire for clarity of evidence, trust in recommendation, and perceived workload reduction (small internal study is acceptable for capstone). ⁶⁴

Evaluation protocols (rigorous, practical)

- **Train/validation/test splits:** fixed split with stratification; for deepfakes include **cross-dataset tests** (train on one dataset, test on another) to measure generalization. ¹⁰⁰
- **Ablation tests:** evaluate contributions of (a) enrichment features, (b) multimodal fusion, (c) correlation rules.
- **Robustness checks:** compression/noise perturbations for deepfakes; domain shift tests for phishing templates; payload obfuscation patterns for web attacks (within dataset constraints). ¹⁰¹
- **Calibration:** report reliability curves or threshold sensitivity for any model used to drive escalation or response recommendation (risk control). ¹⁰²

Risks and ethics (controls and mitigations)

Data privacy

Behavioral biometrics and authentication telemetry can be sensitive. The project will store only derived behavioral features (timing/trajectory statistics) where possible and avoid capturing raw content (typed text) unless already anonymized by the dataset. ⁴²

False actions / unsafe automation

SentinelAI will implement **human-in-the-loop** approval for high-impact actions (e.g., account lock, endpoint isolation). Automated actions, if any, will be limited to low-risk steps (ticket creation, tagging, enrichment) and will be fully logged for audit. ¹⁰³

Deepfake misuse risk

Deepfake detection research can be dual-use. The group project will focus on detection and evaluation. No tooling for creating impersonation content is required for the deliverables beyond use of standard public datasets and published benchmarks. ¹⁰⁴

Bias, drift, and trust

Models that drive triage decisions must be monitored for drift and tested across conditions. The project will document dataset limitations and perform cross-domain tests where feasible, aligning with AI risk management good practices. ¹⁰⁵

Resources, Expected Deliverables, and References

Resources and feasibility

Compute: Deepfake training may require GPU; if constrained, the project can use (a) pretrained backbones and (b) reduced frame sampling while still performing meaningful evaluation on benchmark subsets. ¹⁰⁶

Tooling feasibility:

- SOC simulation: Caldera + Attack Range enable controlled lab telemetry generation with defensible documentation. ⁶⁵

- Data standards: OCSF + ATT&CK STIX enable consistent normalization and tagging without vendor lock-in. ⁶⁷

Expected deliverables (group v2.0 → full capstone deliverables)

- 1) **Group Research Report (this document) + final capstone report** with experiments, tables, diagrams, and conclusions.
- 2) **Four module prototypes** (phishing, CA, web attacks, deepfake) each runnable on a public dataset.
- 3) **SentinelAI integration prototype:** event normalization, correlation, incident view, and recommended response actions (guardrailed).
- 4) **Datasets & experiment scripts** (download/preprocess pipelines, evaluation harness).
- 5) **Presentation deck outline (10–12 slides)** and final Google Slides.
- 6) **Demo scenario** demonstrating multi-signal incident correlation (generated logs). ⁷⁷

Slide deck outline (10–12 slides)

Slide 1: Title, team members, problem statement (multi-domain threats + SOC overload) ⁷⁶

Slide 2: Motivation with evidence (APWG phishing trends; SOC alert fatigue sources) ¹⁰⁷

Slide 3: Group objective and scope (SMART objectives O1–O5)

Slide 4: Literature snapshot (5–8 key sources per module; why chosen) ¹⁰⁸

Slide 5: Overall ecosystem architecture (one diagram) ⁷

Slide 6: Module A — phishing: datasets + model pipeline ¹⁰⁹

Slide 7: Module B — continuous auth + web attacks: datasets + model pipeline ¹¹⁰

Slide 8: Module C — deepfake: datasets + multimodal approach ¹²

Slide 9: Module D — SentinelAI: correlation/risk scoring/response safeguards ¹¹¹

Slide 10: Evaluation plan (metrics + protocols + cross-dataset tests) ¹¹²

Slide 11: Risks & ethics (privacy, human-in-loop) ¹¹³

Slide 12: Timeline + expected deliverables (Gantt summary + demo plan) ¹¹⁴

References (IEEE style) with primary links

(To paste into Google Docs, keep as-is. URLs are provided inside a code block to remain policy-compliant while still being copyable/clickable in your doc.)

[1] Anti-Phishing Working Group (APWG), “Phishing Activity Trends Report, 4th Quarter 2024,” 2025.

https://docs.apwg.org/reports/apwg_trends_report_q4_2024.pdf

- [2] PhishTank, "PhishTank: Collaborative clearing house for phishing data (API)," accessed 2026-03-11.
<https://www.phishtank.com/>
https://phishtank.net/api_info.php
- [3] Carnegie Mellon University, "Enron Email Dataset," accessed 2026-03-11.
<https://www.cs.cmu.edu/~enron/>
- [4] G. Xiang, J. Hong, C. P. Rose, and L. Cranor, "CANTINA+: A Feature-Rich Machine Learning Framework for Detecting Phishing Web Sites," ACM, 2011.
<https://dl.acm.org/doi/10.1145/2019599.2019606>
- [5] A. Aljofey et al., "BERT-PhishFinder: A Robust Model for Accurate Phishing Detection," IEEE Computer Society, 2025.
<https://www.computer.org/csdl/journal/tq/2025/04/10904020>
- [6] L. Halgas, "Detecting Phishing Attacks using Recurrent Neural Networks," 2020.
<https://ora.ox.ac.uk/objects/uuid:8414084e-fa10-4400-9783-b7783082c12a/files/m577acff055854b6ff5983ec9a79380d1>
- [7] G. Vrbančič et al., "Datasets for phishing websites detection," Data in Brief, 2020.
<https://www.sciencedirect.com/science/article/pii/S2352340920313202>
- [8] "PhreshPhish: A Real-World, High-Quality, Large-Scale Phishing URL Dataset," arXiv, 2025.
<https://arxiv.org/html/2507.10854v1>
- [9] K. S. Killourhy and R. A. Maxion, "Comparing Anomaly-Detection Algorithms for Keystroke Dynamics," DSN 2009.
<https://www.cs.cmu.edu/~maxion/pubs/KillourhyMaxion09.pdf>
- [10] S. Eberz et al., "Evaluating Behavioral Biometrics for Continuous Authentication: Challenges and Metrics," AsiaCCS 2017.
<https://lenders.ch/publications/conferences/asiaccs17.pdf>
- [11] M. Frank et al., "Touchalytics: On the Applicability of Touchscreen Input as a Behavioral Biometric for Continuous Authentication," 2012.
<https://arxiv.org/abs/1207.6231>
<https://www.mariofrank.net/paper/touchalytics.pdf>
- [12] Z. Sitová et al., "HMOG: A New Biometric Modality for Continuous Authentication of Smartphone Users," arXiv, 2015.
<https://arxiv.org/abs/1501.01199>
Dataset repo: <https://github.com/hmog-dataset/hmog>

- [13] G. Stragapede et al., "BehavePassDB: Public Database for Mobile Behavioral Biometrics and Benchmark Evaluation," Pattern Recognition, 2023.
<https://doi.org/10.1016/j.patcog.2022.109089>
<https://arxiv.org/abs/2206.02502>
- [14] J. Liang et al., "Anomaly-Based Web Attack Detection: A Deep Learning Approach," ACM, 2017.
<https://dl.acm.org/doi/10.1145/3171592.3171594>
- [15] J. Wang et al., "Evaluating CNN and LSTM for Web Attack Detection," ACM, 2018.
<https://dl.acm.org/doi/10.1145/3195106.3195107>
- [16] Z. Tian et al., "A Distributed Deep Learning System for Web Attack Detection on Edge Devices," IEEE TII, doi:10.1109/TII.2019.2938778, 2019.
<https://doi.org/10.1109/TII.2019.2938778>
- [17] A. Tekerek, "A Novel Architecture for Web-Based Attack Detection Using Convolutional Neural Network," Computers & Security, 2021.
<https://www.sciencedirect.com/science/article/abs/pii/S0167404820303692>
- [18] T. S. Riera et al., "A new multi-label dataset for Web attacks CAPEC classification using machine learning techniques," Computers & Security, 2022.
<https://doi.org/10.1016/j.cose.2022.102788>
- [19] A. Rössler et al., "FaceForensics++: Learning to Detect Manipulated Facial Images," ICCV 2019.
https://openaccess.thecvf.com/content_ICCV_2019/papers/Rossler_FaceForensics_Learning_to_Detect_Manipulated_Facial_Images_ICCV_2019_paper.pdf
- [20] B. Dolhansky et al., "The DeepFake Detection Challenge (DFDC) Dataset," arXiv, 2020.
<https://arxiv.org/abs/2006.07397>
 Dataset page: <https://ai.meta.com/datasets/dfdc/>
- [21] Y. Li et al., "Celeb-DF: A Large-Scale Challenging Dataset for DeepFake Forensics," CVPR 2020.
https://openaccess.thecvf.com/content_CVPR_2020/papers/Li_Celeb-DF_A_Large-Scale_Challenging_Dataset_for_DeepFake_Forensics_CVPR_2020_paper.pdf
- [22] L. Jiang et al., "DeeperForensics-1.0: A Large-Scale Dataset for Real-World Face Forgery Detection," CVPR 2020.
https://openaccess.thecvf.com/content_CVPR_2020/papers/Jiang_DeeperForensics-1.0_A_Large-Scale_Dataset_for_Real-World_Face_Forgery_Detection_CVPR_2020_paper.pdf
- [23] A. Haliassos et al., "Lips Don't Lie: A Generalisable and Robust Approach

To Face Forgery Detection," CVPR 2021.

https://openaccess.thecvf.com/content/CVPR2021/papers/Haliassos_Lips_Dont_Lie_A_Generalisable_and_Robust_Approach_To_Face_CVPR_2021_paper.pdf

[24] J. S. Chung and A. Zisserman, "Out of Time: Automated Lip Sync in the Wild," 2016.

<https://www.robots.ox.ac.uk/~vgg/publications/2016/Chung16a/chung16a.pdf>

[25] NIST, "SP 800-61r3: Incident Response Recommendations and Considerations for Cybersecurity Risk Management," 2025.

<https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-61r3.pdf>

[26] NIST, "SP 800-92: Guide to Computer Security Log Management," 2006.

<https://nvlpubs.nist.gov/nistpubs/legacy/sp/nistspecialpublication800-92.pdf>

[27] NIST, "SP 800-137: Information Security Continuous Monitoring (ISCM)," 2011.

<https://nvlpubs.nist.gov/nistpubs/legacy/sp/nistspecialpublication800-137.pdf>

[28] MITRE, "MITRE ATT&CK: Design and Philosophy," 2020.

https://attack.mitre.org/docs/ATTACK_Design_and_Philosophy_March_2020.pdf

ATT&CK site: <https://attack.mitre.org/>

ATT&CK STIX: <https://attack.mitre.org/resources/attack-data-and-tools/>

[29] Open Cybersecurity Schema Framework (OCSF), "OCSF: vendor-agnostic core security schema," accessed 2026-03-11.

<https://ocsf.io/>

<https://github.com/ocsf/ocsf-schema>

[30] MITRE, "Caldera: adversary emulation platform leveraging ATT&CK," accessed 2026-03-11.

<https://caldera.readthedocs.io/>

<https://www.mitre.org/ourimpact/intellectual-property/caldera>

[31] Splunk, "Splunk Attack Range: instrumented environments + attack simulation for detection engineering," accessed 2026-03-11.

https://github.com/splunk/attack_range

[32] K. Peffers et al., "A Design Science Research Methodology for Information Systems Research," JMIS, 2007-2008.

<https://indico.cern.ch/event/1542774/contributions/6494311/attachments/3080345/5465431/>

[Peffers_2007_A%20Design%20Science%20Research%20Methodology%20for%20Information%20Systems%20Research](#)

[33] A. Hevner et al., "Design Science in Information Systems Research," MIS Quarterly, 2004.

<https://www.uni-kassel.de/fb07/index.php?eID=dumpFile&f=4899&t=f&token=fda52302d42e5e5c05d7f89c2d578f80fedf7b28>

[34] NIST, "AI Risk Management Framework (AI RMF 1.0)," 2023.
<https://nvlpubs.nist.gov/nistpubs/ai/nist.ai.100-1.pdf>

[35] ACM, "Alert Fatigue in Security Operations Centres," 2025.
<https://dl.acm.org/doi/10.1145/3723158>

[36] X. Wang et al., "Combating alert fatigue with AlertPro: Context-aware alert prioritization ...," 2024.
<https://www.sciencedirect.com/science/article/abs/pii/S0167404823004935>

1 16 64 76 <https://dl.acm.org/doi/10.1145/3723158>

<https://dl.acm.org/doi/10.1145/3723158>

2 27 71 109 <https://www.phishtank.com/>

<https://www.phishtank.com/>

3 21 37 91 <https://www.cs.cmu.edu/~maxion/pubs/KillourhyMaxion09.pdf>

<https://www.cs.cmu.edu/~maxion/pubs/KillourhyMaxion09.pdf>

4 10 20 24 44 92 98 108 <https://dl.acm.org/doi/10.1145/3171592.3171594>

<https://dl.acm.org/doi/10.1145/3171592.3171594>

5 12 22 51 74 82 101 **Learning to Detect Manipulated Facial Images**

[https://openaccess.thecvf.com/content_ICCV_2019/papers/](https://openaccess.thecvf.com/content_ICCV_2019/papers/Rossler_FaceForensics_Learning_to_Detect_Manipulated_Facial_Images_ICCV_2019_paper.pdf?utm_source=chatgpt.com)

[Rossler_FaceForensics_Learning_to_Detect_Manipulated_Facial_Images_ICCV_2019_paper.pdf?utm_source=chatgpt.com](https://openaccess.thecvf.com/content_ICCV_2019/papers/Rossler_FaceForensics_Learning_to_Detect_Manipulated_Facial_Images_ICCV_2019_paper.pdf?utm_source=chatgpt.com)

6 13 14 17 59 68 69 70 97 103 111 [https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.](https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-61r3.pdf)

800-61r3.pdf

<https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-61r3.pdf>

7 25 66 67 96 <https://ocsf.io/>

<https://ocsf.io/>

8 26 34 36 42 72 78 90 99 110 112 <https://lenders.ch/publications/conferences/asiaccs17.pdf>

<https://lenders.ch/publications/conferences/asiaccs17.pdf>

9 107 https://docs.apwg.org/reports/apwg_trends_report_q4_2024.pdf

https://docs.apwg.org/reports/apwg_trends_report_q4_2024.pdf

11 35 **A Review of Continuous Authentication Using Behavioral ...**

https://dl.acm.org/doi/10.1145/2984393.2984403?utm_source=chatgpt.com

15 52 104 106 <https://arxiv.org/abs/2006.07397>

<https://arxiv.org/abs/2006.07397>

18 83 <https://www.unb.ca/cic/datasets/ids-2017.html>

<https://www.unb.ca/cic/datasets/ids-2017.html>

19 32 Data Article Datasets for phishing websites detection

https://www.sciencedirect.com/science/article/pii/S2352340920313202?utm_source=chatgpt.com

23 65 77 87 <https://caldera.readthedocs.io/>

<https://caldera.readthedocs.io/>

28 80 <https://www.cs.cmu.edu/~enron/>

<https://www.cs.cmu.edu/~enron/>

29 <https://dl.acm.org/doi/10.1145/2019599.2019606>

<https://dl.acm.org/doi/10.1145/2019599.2019606>

30 89 BERT-PhishFinder: A Robust Model for Accurate Phishing ...

https://www.computer.org/csdl/journal/tq/2025/04/10904020/24CDGDy1EWs?utm_source=chatgpt.com

31 <https://ora.ox.ac.uk/objects/uuid%3A8414084e-fa10-4400-9783-b7783082c12a/files/m577acff055854b6ff5983ec9a79380d1>

<https://ora.ox.ac.uk/objects/uuid%3A8414084e-fa10-4400-9783-b7783082c12a/files/m577acff055854b6ff5983ec9a79380d1>

33 PhreshPhish: A Real-World, High-Quality, Large-Scale ...

https://arxiv.org/html/2507.10854v1?utm_source=chatgpt.com

38 <https://github.com/balabit/Mouse-Dynamics-Challenge>

<https://github.com/balabit/Mouse-Dynamics-Challenge>

39 <https://www.mariofrank.net/paper/touchalytics.pdf>

<https://www.mariofrank.net/paper/touchalytics.pdf>

40 <https://arxiv.org/abs/1501.01199>

<https://arxiv.org/abs/1501.01199>

41 <https://www.sciencedirect.com/science/article/pii/S0031320322005696>

<https://www.sciencedirect.com/science/article/pii/S0031320322005696>

43 73 HTTP DATASET CSIC 2010 What is this?

https://petescully.co.uk/wp-content/uploads/2018/04/http_dataset_csic_2010.pdf?utm_source=chatgpt.com

45 <https://dl.acm.org/doi/10.1145/3195106.3195107>

<https://dl.acm.org/doi/10.1145/3195106.3195107>

46 <https://syslog.co.in/wp-content/uploads/2019/11/A-Distributed-Deep-Learning-System-for-Web-Attack-Detection-on-Edge-Devices.pdf>

<https://syslog.co.in/wp-content/uploads/2019/11/A-Distributed-Deep-Learning-System-for-Web-Attack-Detection-on-Edge-Devices.pdf>

47 <https://www.sciencedirect.com/science/article/abs/pii/S0167404820303692>

<https://www.sciencedirect.com/science/article/abs/pii/S0167404820303692>

48 50 93 <https://dl.acm.org/doi/10.1016/j.cose.2022.102788>

<https://dl.acm.org/doi/10.1016/j.cose.2022.102788>

49 <https://attack.mitre.org/resources/attack-data-and-tools/>

<https://attack.mitre.org/resources/attack-data-and-tools/>

- 53 https://openaccess.thecvf.com/content_CVPR_2020/papers/Li_Celeb-DF_A_Large-Scale_Challenging_Dataset_for_DeepFake_Forensics_CVPR_2020_paper.pdf
https://openaccess.thecvf.com/content_CVPR_2020/papers/Li_Celeb-DF_A_Large-Scale_Challenging_Dataset_for_DeepFake_Forensics_CVPR_2020_paper.pdf
- 54 https://openaccess.thecvf.com/content_CVPR_2020/papers/Jiang_DeepForensics-1.0_A_Large-Scale_Dataset_for_Real-World_Face_Forgery_Detection_CVPR_2020_paper.pdf
https://openaccess.thecvf.com/content_CVPR_2020/papers/Jiang_DeepForensics-1.0_A_Large-Scale_Dataset_for_Real-World_Face_Forgery_Detection_CVPR_2020_paper.pdf
- 55 58 94 100 https://openaccess.thecvf.com/content/CVPR2021/papers/Haliassos_Lips_Dont_Lie_A_Generalisable_and_Robust_Approach_To_Face_CVPR_2021_paper.pdf
https://openaccess.thecvf.com/content/CVPR2021/papers/Haliassos_Lips_Dont_Lie_A_Generalisable_and_Robust_Approach_To_Face_CVPR_2021_paper.pdf
- 56 95 <https://www.robots.ox.ac.uk/~vgg/publications/2016/Chung16a/chung16a.pdf>
<https://www.robots.ox.ac.uk/~vgg/publications/2016/Chung16a/chung16a.pdf>
- 57 <https://farid.berkeley.edu/downloads/publications/cvprw24c.pdf>
<https://farid.berkeley.edu/downloads/publications/cvprw24c.pdf>
- 60 88 <https://nvlpubs.nist.gov/nistpubs/legacy/sp/nistspecialpublication800-92.pdf>
<https://nvlpubs.nist.gov/nistpubs/legacy/sp/nistspecialpublication800-92.pdf>
- 61 <https://nvlpubs.nist.gov/nistpubs/legacy/sp/nistspecialpublication800-137.pdf>
<https://nvlpubs.nist.gov/nistpubs/legacy/sp/nistspecialpublication800-137.pdf>
- 62 Design and Philosophy
https://attack.mitre.org/docs/ATTACK_Design_and_Philosophy_March_2020.pdf?utm_source=chatgpt.com
- 63 SoK: The MITRE ATT&CK Framework in Research and Practice
https://arxiv.org/abs/2304.07411?utm_source=chatgpt.com
- 75 <https://www.uni-kassel.de/fb07/index.php?eID=dumpFile&f=4899&t=f&token=fda52302d42e5e5c05d7f89c2d578f80fedf7b28>
<https://www.uni-kassel.de/fb07/index.php?eID=dumpFile&f=4899&t=f&token=fda52302d42e5e5c05d7f89c2d578f80fedf7b28>
- 79 102 105 113 <https://nvlpubs.nist.gov/nistpubs/ai/nist.ai.100-1.pdf>
<https://nvlpubs.nist.gov/nistpubs/ai/nist.ai.100-1.pdf>
- 81 <https://www.cs.cmu.edu/~keystroke/>
<https://www.cs.cmu.edu/~keystroke/>
- 84 <https://csr.lanl.gov/data/auth/>
<https://csr.lanl.gov/data/auth/>
- 85 <https://www.sei.cmu.edu/library/insider-threat-test-dataset/>
<https://www.sei.cmu.edu/library/insider-threat-test-dataset/>
- 86 114 https://github.com/splunk/attack_range
https://github.com/splunk/attack_range